



Sadhana Devi S V 250701620-2025-CSE ▾

S2**Started on** Monday, 10 August 2026, 4:01 PM**State** Finished**Completed on** Monday, 10 August 2026, 4:09 PM**Time taken** 7 mins 3 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int i = 1;
7      int s = 1;
8      int count = 2;
9
10     scanf("%d", &n);
11
12     while (s <= n)
13     {
14         count += 3;
15         i++;
16         s += i;
17     }
18
19     count++;
20
21     printf("%d", count);
22
23     return 0;
24 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 10 August 2026, 4:09 PM**State** Finished**Completed on** Monday, 10 August 2026, 4:10 PM**Time taken** 1 min 14 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int count = 0;
7
8      scanf("%d", &n);
9
10     count++;
11
12     if (n == 1)
13     {
14         count++;
15         printf("%d", count);
16     }
17     else
18     {
19         for (int i = 1; i <= n; i++)
20         {
21             count++;
22
23             for (int j = 1; j <= n; j++)
24             {
25                 count++;
26                 count++;
27                 count++;
28                 break;
29             }
30
31             count++;
32         }
33
34         count++;
35         printf("%d", count);
```

```
36 |     }  
37 |  
38 |     return 0;  
39 | }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 10 August 2026, 4:10 PM**State** Finished**Completed on** Monday, 10 August 2026, 4:11 PM**Time taken** 1 min 9 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int num;
6      int i;
7      int count = 0;
8
9      scanf("%d", &num);
10
11     for (i = 1; i <= num; i++)
12     {
13         count++;
14
15         if (num % i == 0)
16         {
17             count++;
18         }
19     }
20
21     count++;
22 }
23
24 count++;
25
26 printf("%d", count);
27
28 return 0;
29 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 10 August 2026, 4:11 PM**State** Finished**Completed on** Monday, 10 August 2026, 4:21 PM**Time taken** 10 mins 16 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int c = 0;
7      int count = 1;
8
9      scanf("%d", &n);
10
11     count++;
12
13     for (int i = n / 2; i < n; i++)
14     {
15         count++;
16
17         for (int j = 1; j < n; j = 2 * j)
18         {
19             count++;
20
21             for (int k = 1; k < n; k = k * 2)
22             {
23                 count++;
24                 c++;
25                 count++;
26             }
27
28             count++;
29         }
30
31         count++;
32     }
33
34     printf("%d", count);
35
36     return 0;
37 }
38 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 10 August 2026, 4:21 PM**State** Finished**Completed on** Monday, 10 August 2026, 4:24 PM**Time taken** 2 mins 46 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int rev = 0, remainder;
7      int count = 0;
8
9      scanf("%d", &n);
10
11     while (n != 0)
12     {
13         count++;
14         remainder = n % 10;
15         count++;
16         rev = rev * 10 + remainder;
17         count++;
18         n /= 10;
19         count++;
20     }
21
22     count++;
23     count++;
24     count++;
25
26     printf("%d", count);
27
28     return 0;
29 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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